



《人工智能与模式识别》

Artificial Intelligence and Pattern Recognition

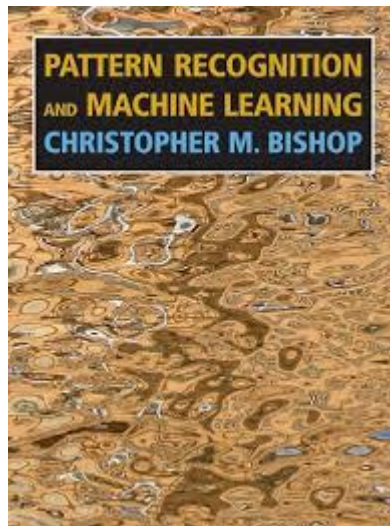
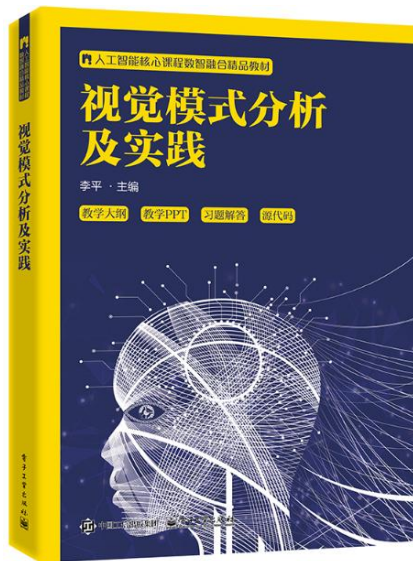
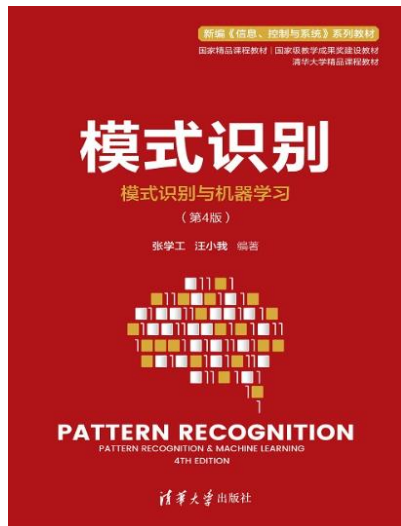
计算机学院
School of Computer Science

Course Overview

- 课程性质：专业学位课
- 课时学分：48学时 / 3学分
- 总评成绩：期末考试60% / 平时表现40%
- 教学形式：讲座、研讨、文献阅读、作业项目等
- 教学平台：学校泛雅平台
- 课程网址：<https://mlvccn.github.io/course-ai/>

Textbook / Reference

- 《模式识别(第四版)》. 张学工, 汪小我. 清华大学出版社.
- 《视觉模式分析及实践》. 李平. 电子工业出版社.
- 《机器学习》. 周志华. 清华大学出版社.
- 《Pattern Recognition and Machine Learning》. Chirstopher M. Bishop, Springer.





WHAT IS A.I.?

AI-1

Artificial
Intelligence

Softwa

AI

Reasoning

Science

Computer

System

Knowledge

Technology

Learning

AI-2

Artificial Narrow Intelligence (ANI)



Stage-1

Machine Learning

- Specialises in one area and solves one problem



Siri



Alexa



Cortana

Artificial General Intelligence (AGI)



Stage-2

Machine Intelligence

- Refers to a computer that is as smart as a human across the board

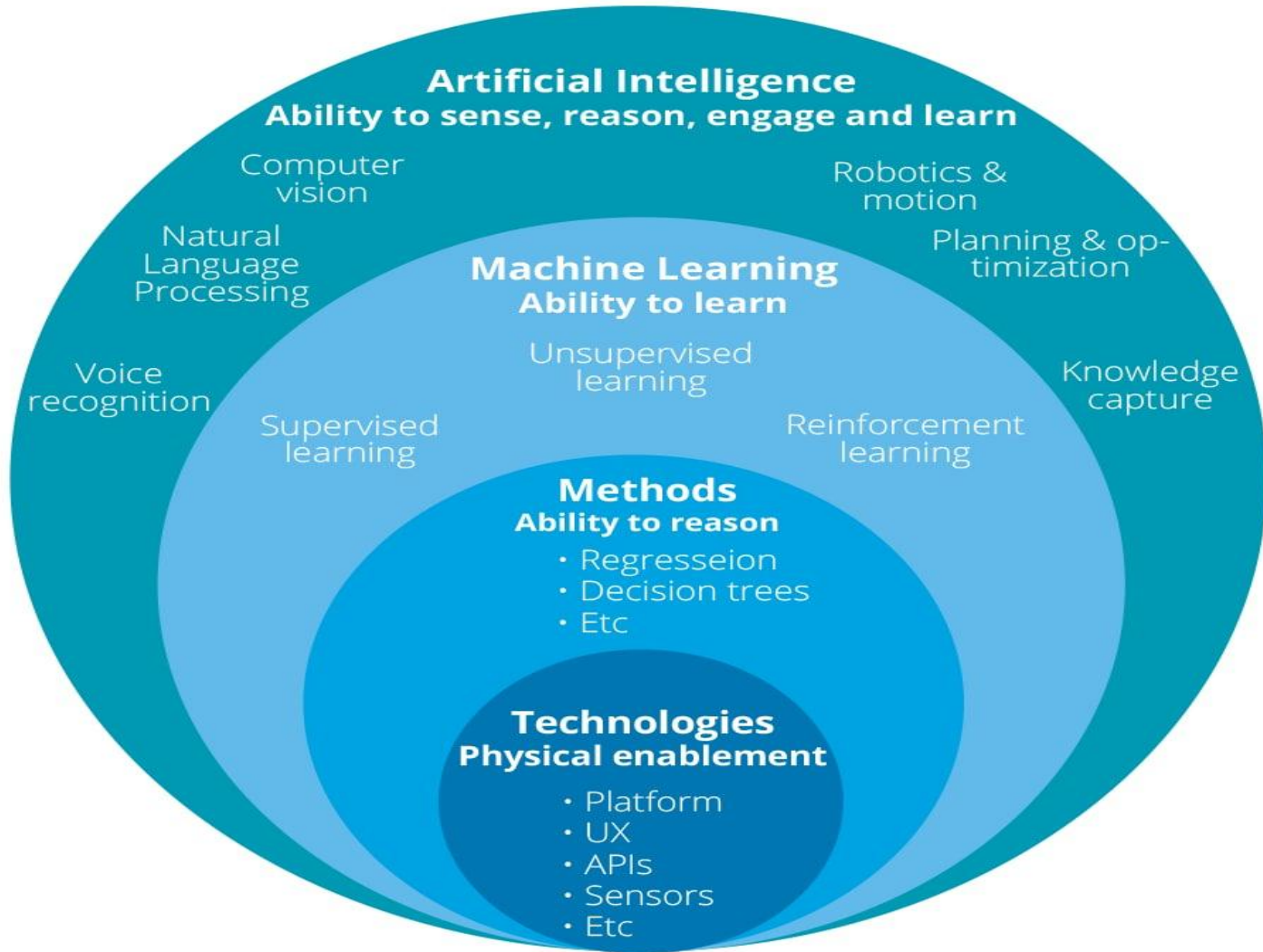
Artificial Super Intelligence (ASI)

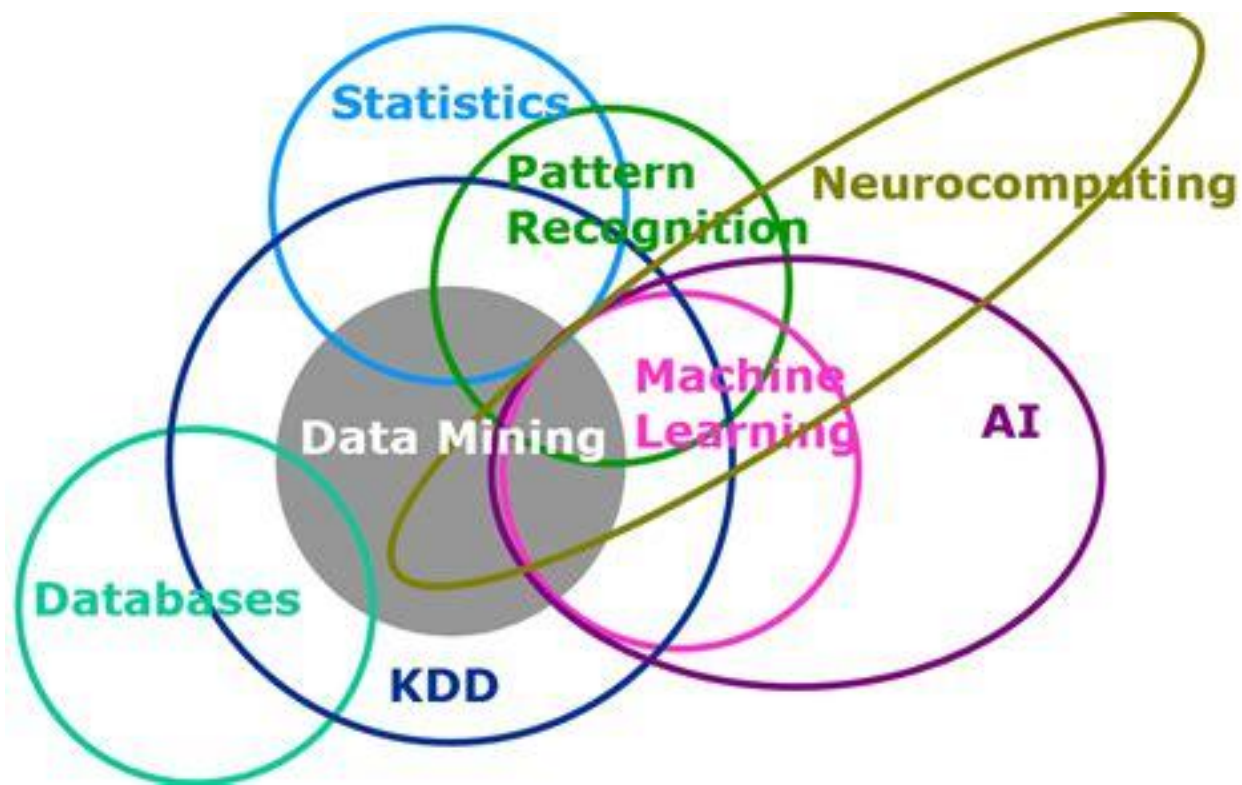


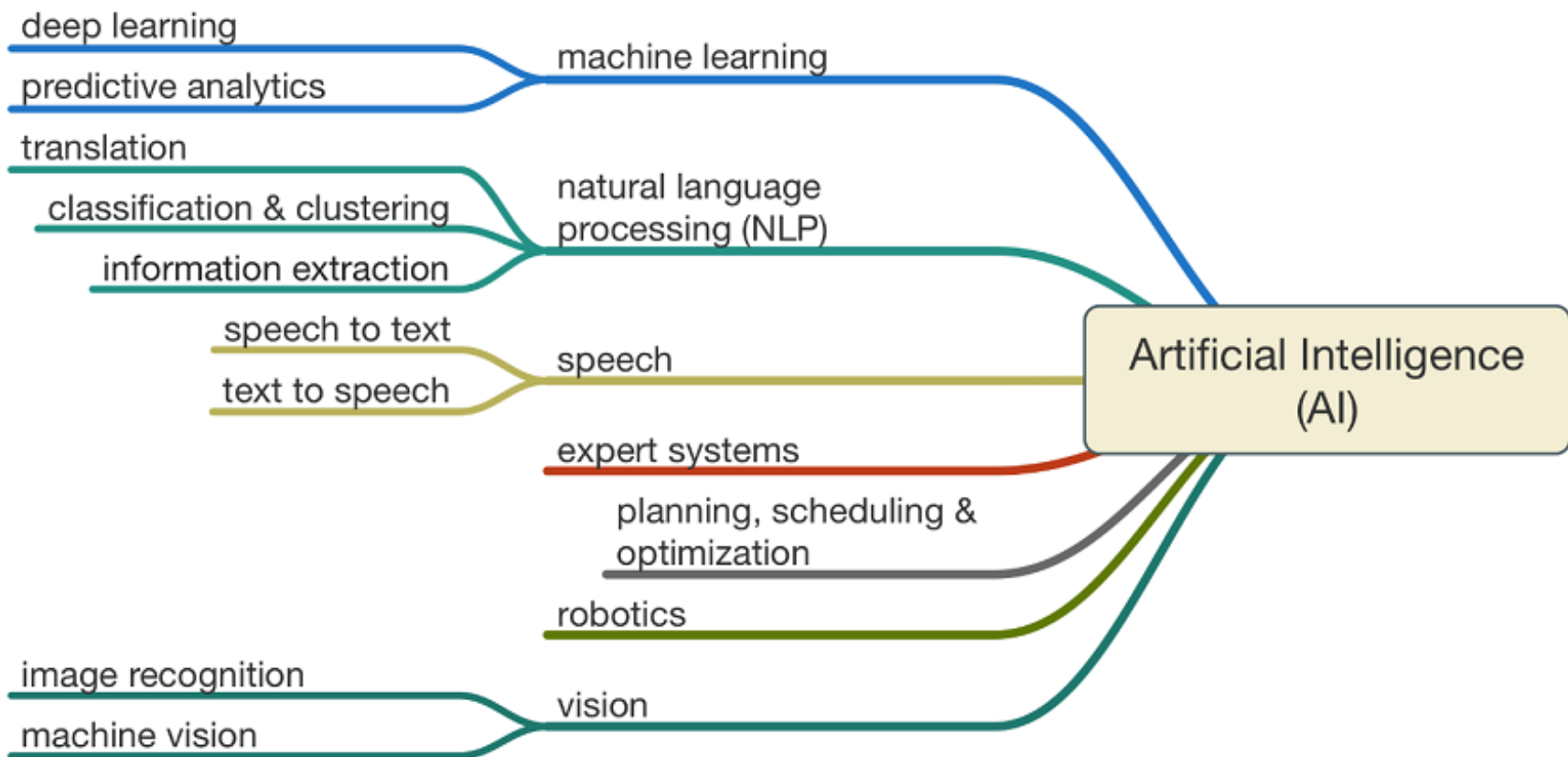
Stage-3

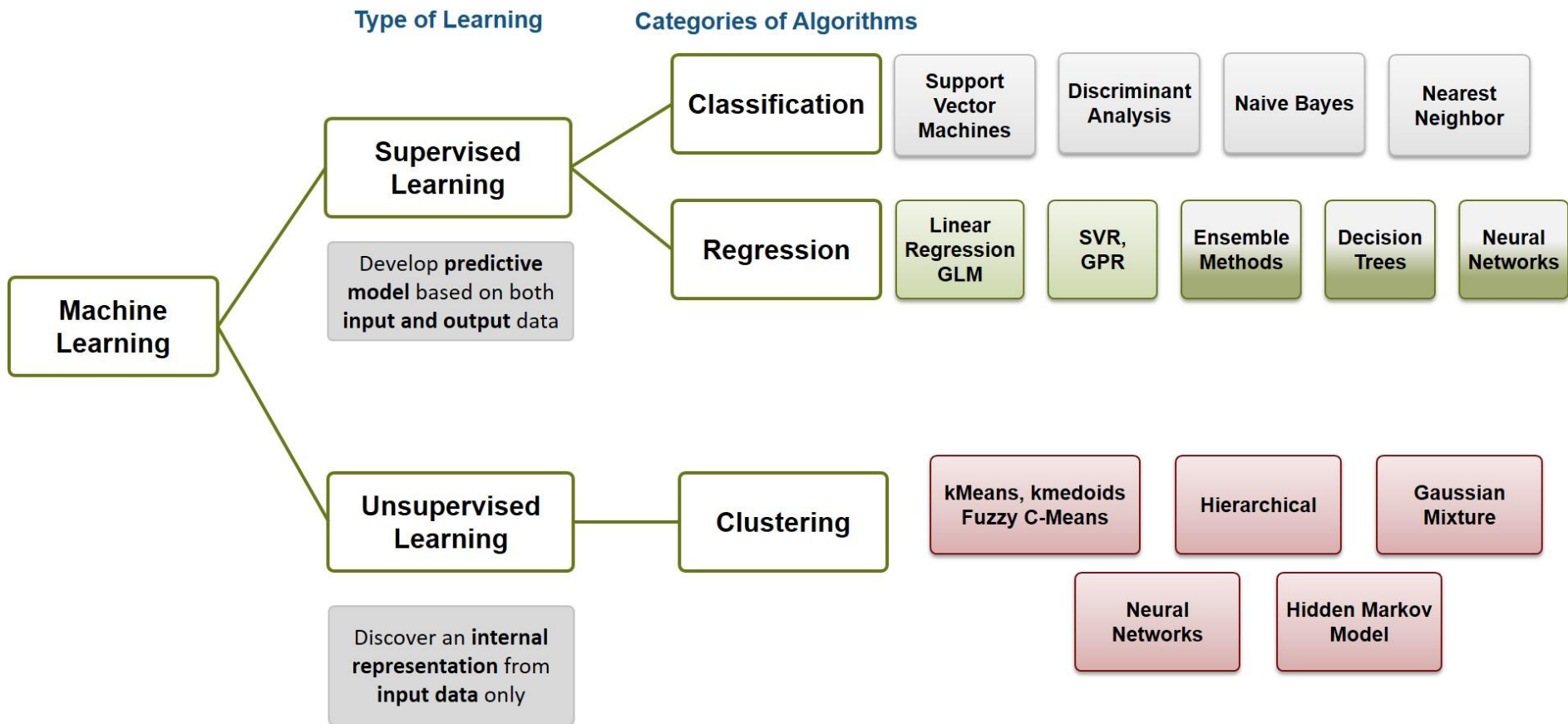
Machine Consciousness

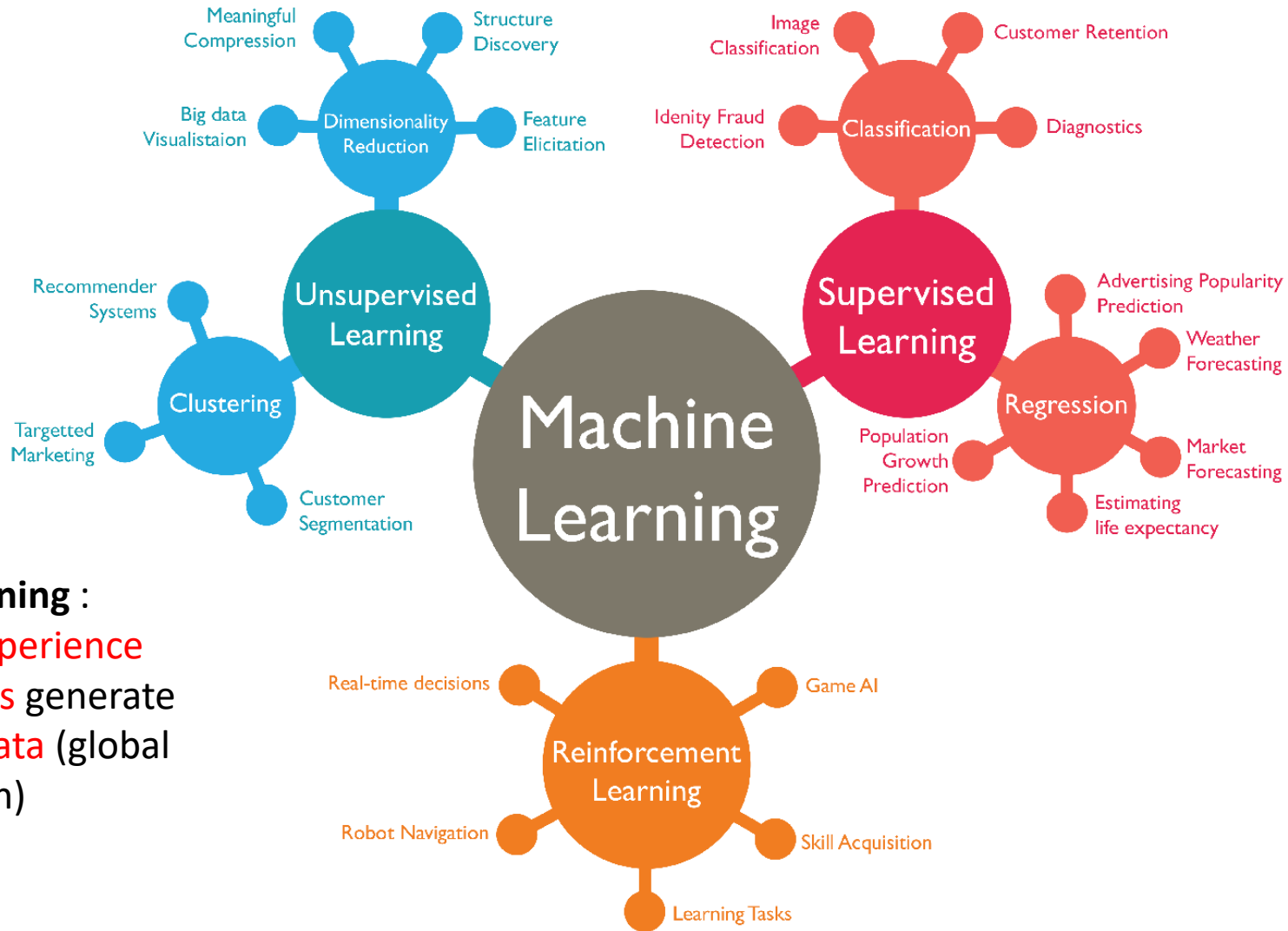
- An intellect that is much smarter than the best human brains in practically every field





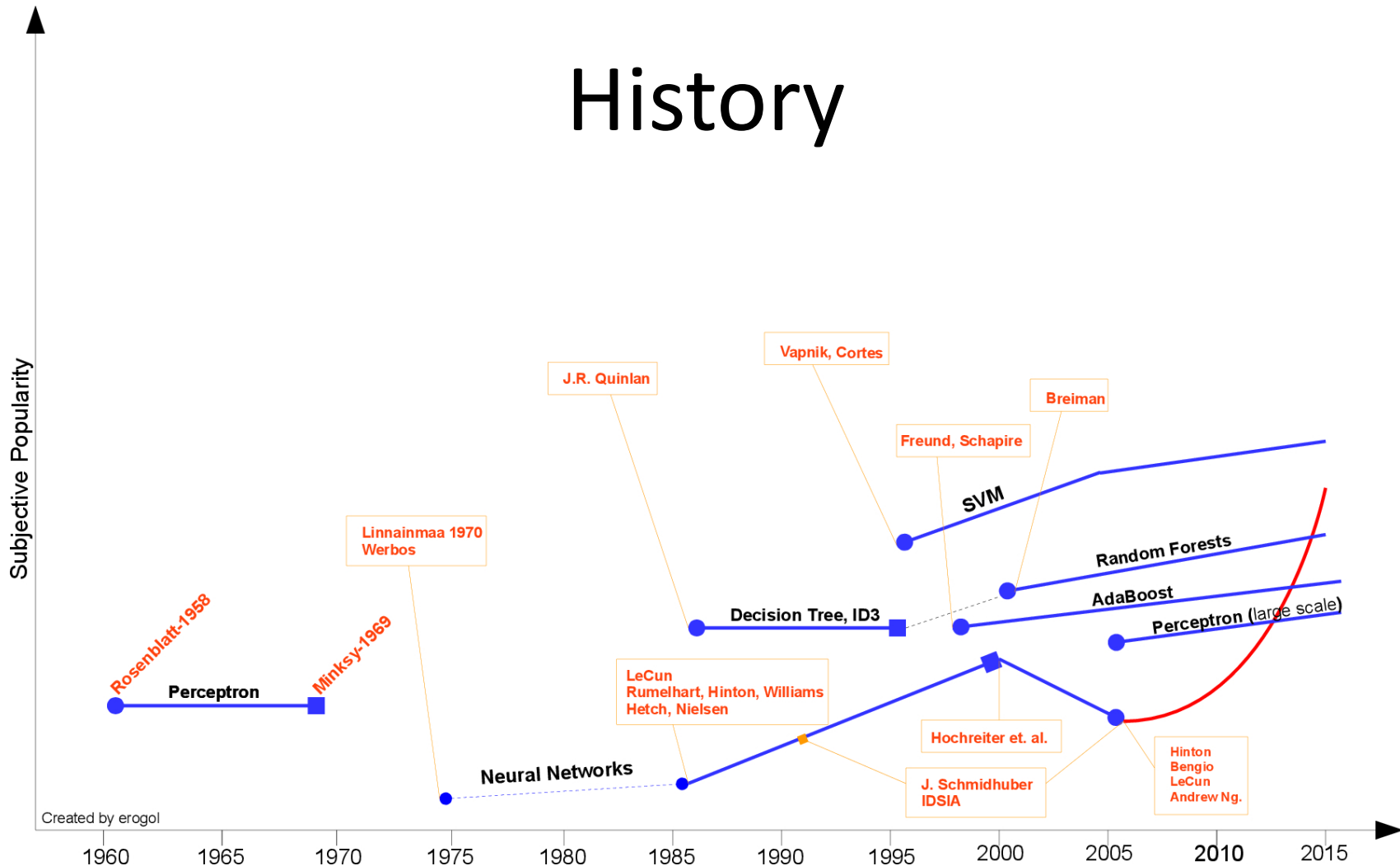






Machine Learning :
Learn from experience
and algorithms generate
model from data (global
& local pattern)

History



Turing Award

- **Leslie Valiant** (Harvard University, 1949-), the founder of computational learning theory, 2010
- **Judea Pearl** (UCLA, 1936-), the pioneer of graphical model learning, 2011
- **Yann LeCun, Geoffrey Hinton** and **Yoshua Bengio**, deep neural networks that reshape how computer systems are built, 2018

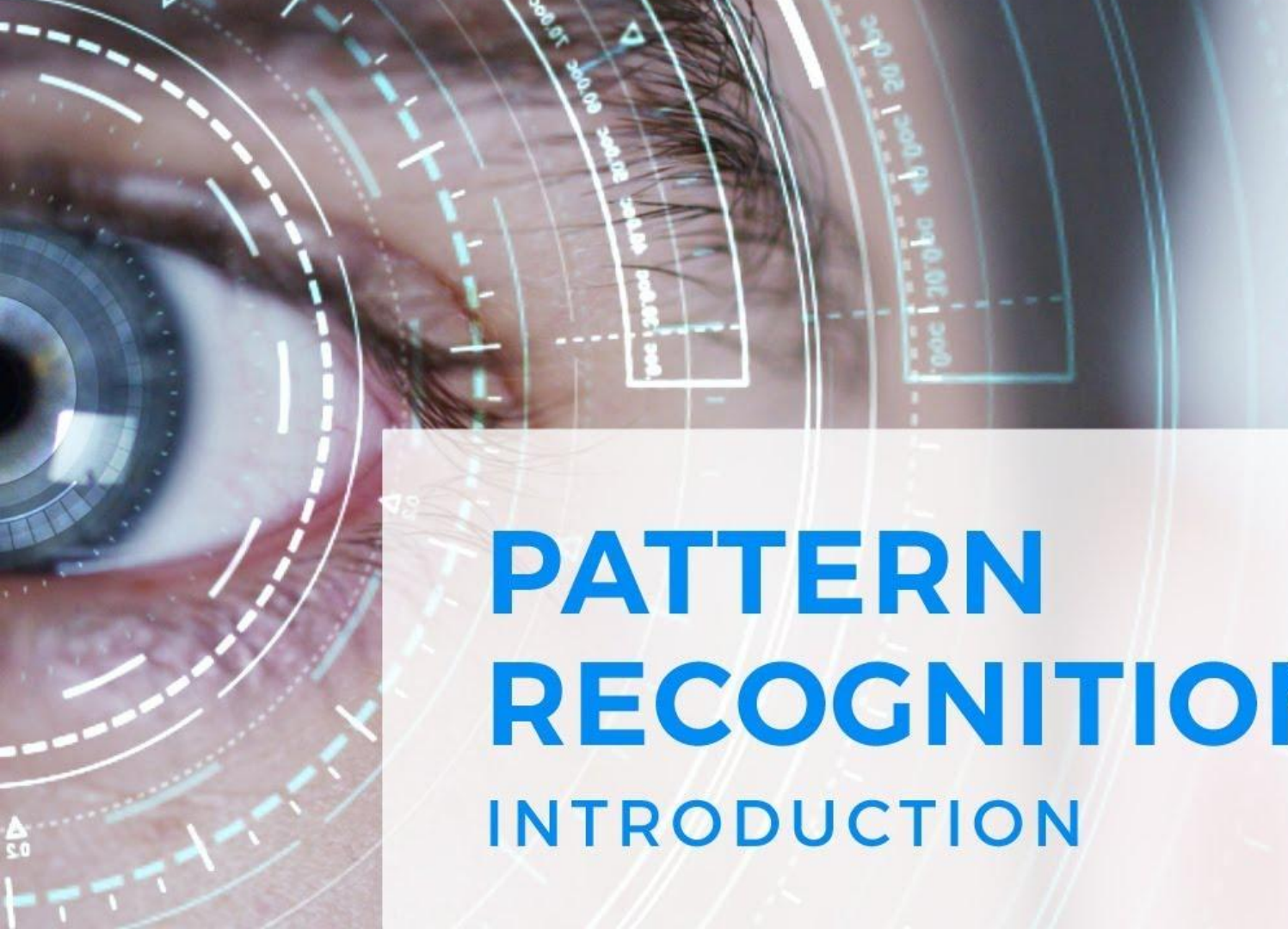
2018 ACM A.M. Turing Award



Yann LeCun: Prof. at New York University, Chief AI Scientist at Facebook

Geoffrey Hinton: Prof. at University of Toronto, Google (*LeCun is his postdoc*)

Yoshua Bengio: Prof. at Université de Montréal of Canada



PATTERN RECOGNITION

INTRODUCTION



Pseudotsuga menziesii



Arbutus menziesii



Acer macrophyllum



06419446010

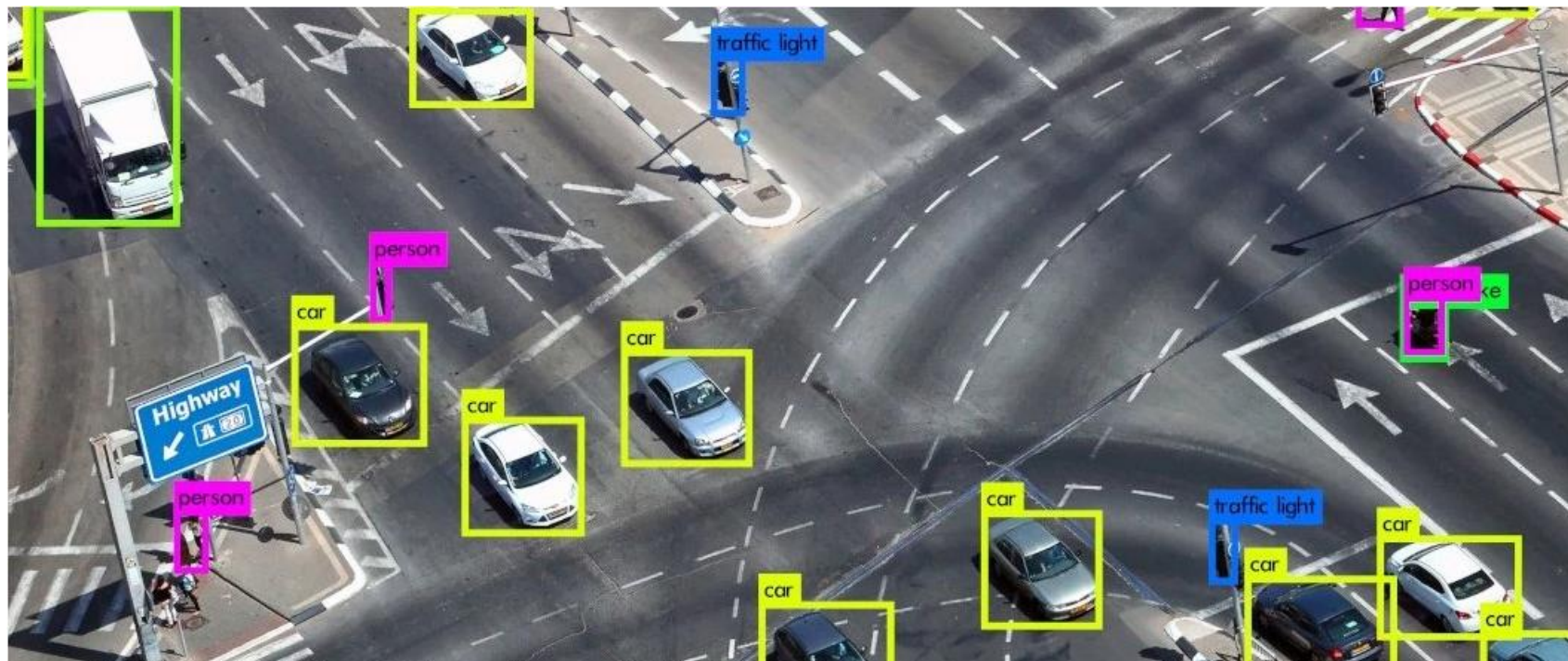


handwriting

福

Patterns include repeated trends in various forms of data.

Pattern is an entity, vaguely defined, and can be any entity of interest that one needs to recognize and identify. One wants to know its name (identity).



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THANK YOU
FOR YOUR BUSINESS

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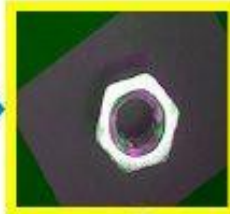
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PART IMAGE



PART IMAGE ANALYSIS



CLASSIFY

GOOD



FAULTY



Pattern recognition is the automated recognition of **patterns and regularities in data**. It has applications in statistical data analysis, signal processing, image analysis, information retrieval, bioinformatics, data compression, computer graphics and machine learning.

Pattern recognition is the process of recognizing **regularities in data** by a machine that **uses machine learning** algorithms. In the heart of the process lies the classification of events based on statistical information, historical data, or the machine's memory.

Pattern recognition is the **use of machine learning algorithms to identify patterns**. It classifies data based on statistical information or knowledge gained from patterns and their representation. It reveals that how machines can observe the environment, and learns to **distinguish various patterns of interest** from their background, identifying the categories of patterns.



Pattern Recognition

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Latest issue

Volume 122

In progress

February 2022

About the journal

Pattern Recognition is a mature field, which underpins developments in cognate fields such as computer vision, image processing, text and document analysis and neural networks. It is closely akin to machine learning, and also finds applications in fast emerging areas such as biometrics, bioinformatics, multimedia data analysis and most recently data science.

[Read more](#)

Pattern Recognition is a mature but exciting and fast developing field, which underpins developments in cognate fields such as **computer vision, image processing, text and document analysis and neural networks**. It is closely akin to **machine learning**, and also finds applications in fast emerging areas such as **biometrics, bioinformatics, multimedia data analysis** and most recently **data science**.

Social Signal Processing

*Socially Perceptive
Computing Lab*

application

inspiration

Pattern Recognition

Pattern Recognition Lab

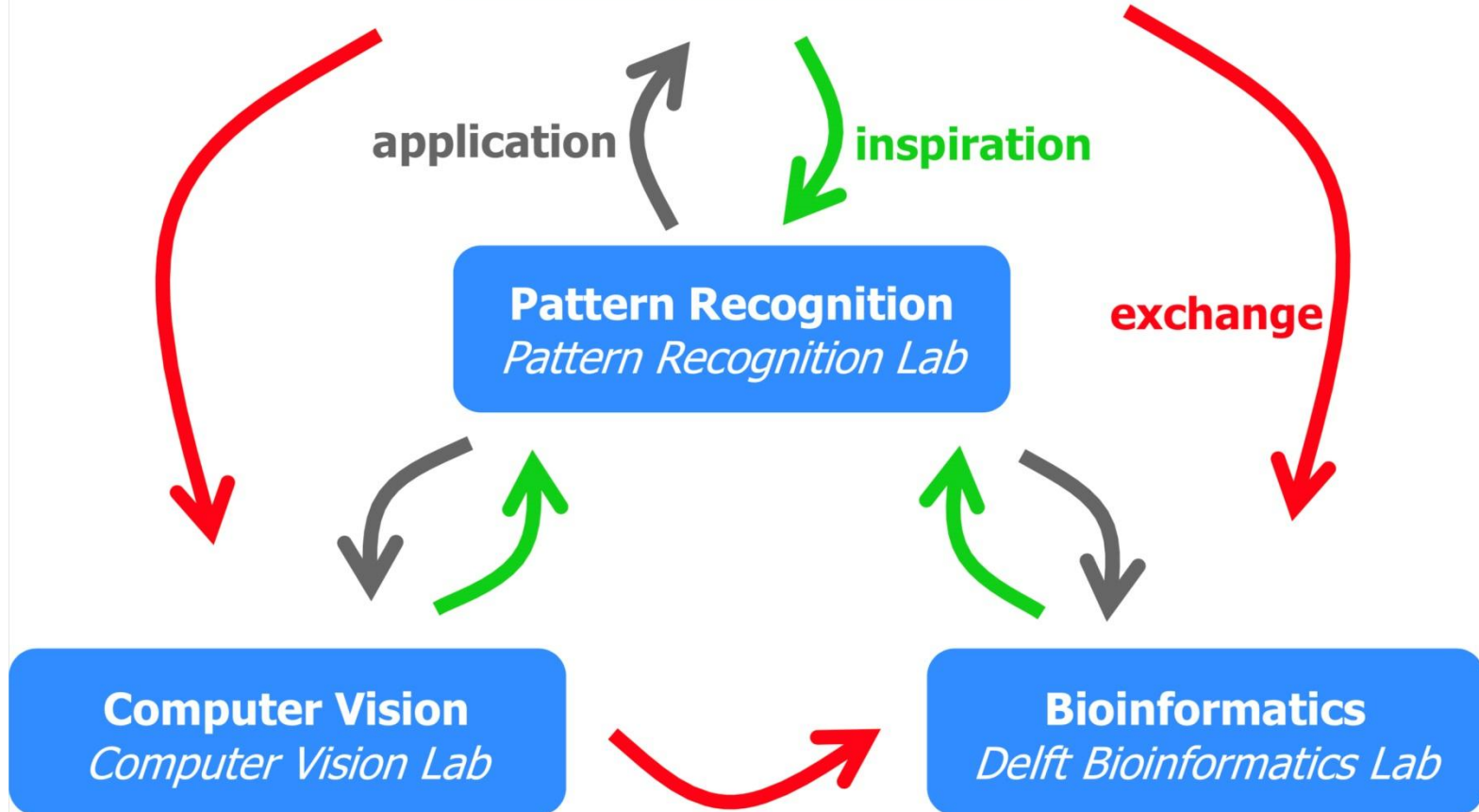
exchange

Computer Vision

Computer Vision Lab

Bioinformatics

Delft Bioinformatics Lab



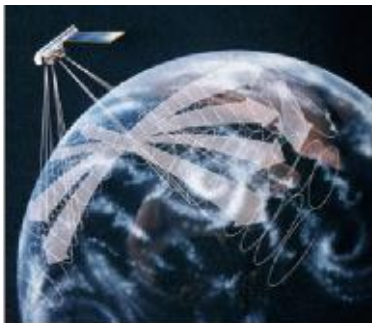


PERFORMATIX
FAST FORWARD INNOVATION



Breaking Down Pattern Recognition: The What, How, and Why

Information
Source
eg. Satellite
Sensor



Information
Preprocessing
eg. Co-Registration

Feature
Extraction

Features

Classifier

Classification
Result
eg. class vector



Green
Yellow
Orange

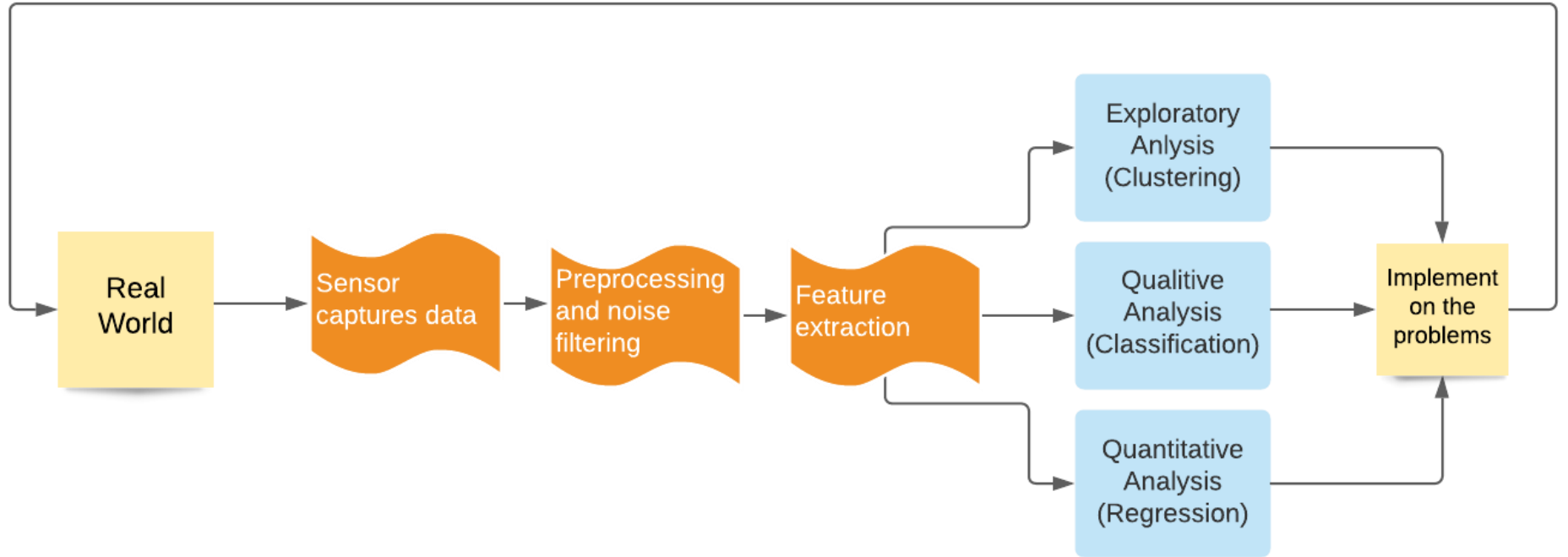
Eyes

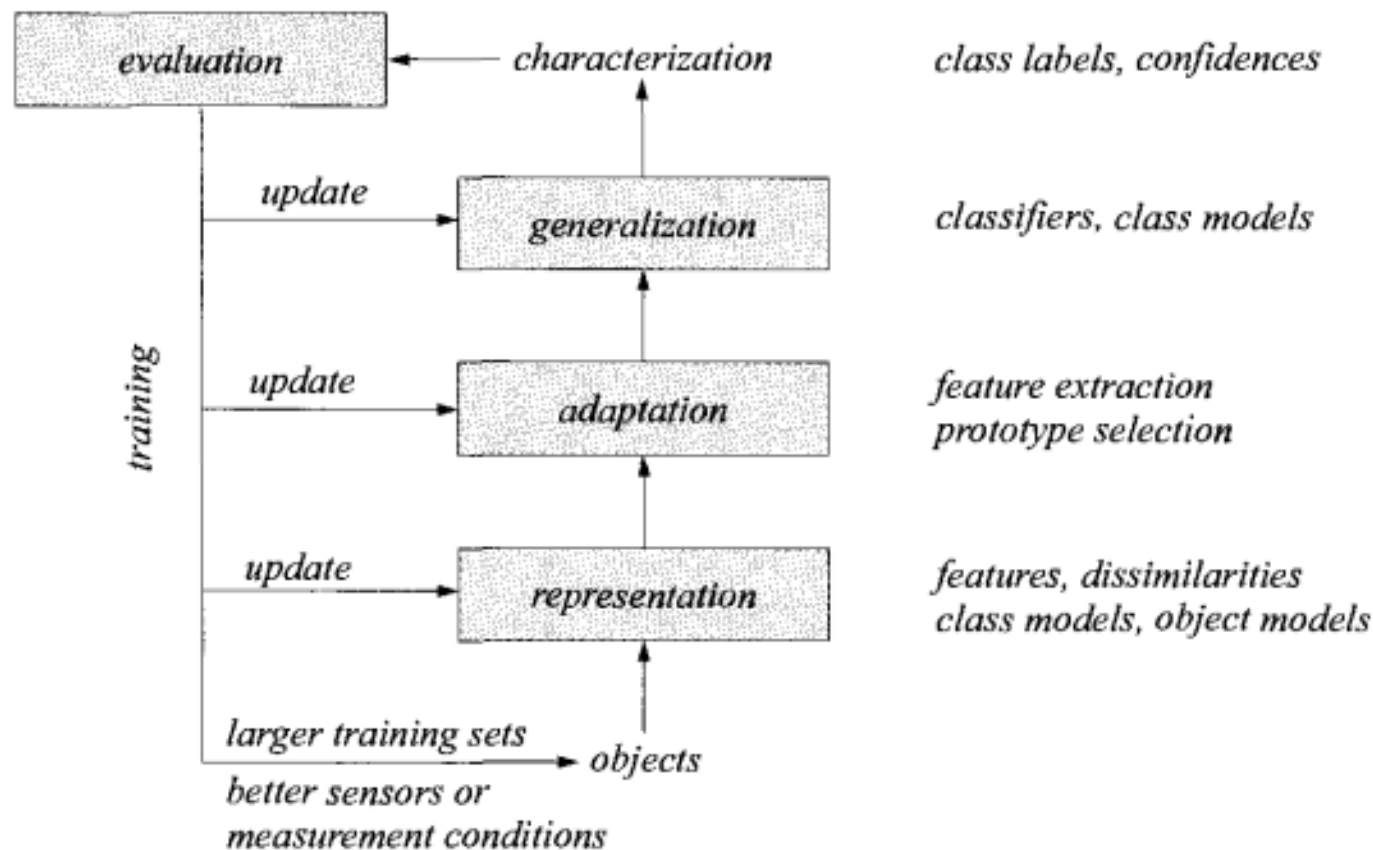
Ginger
Black
Tabby

Fur

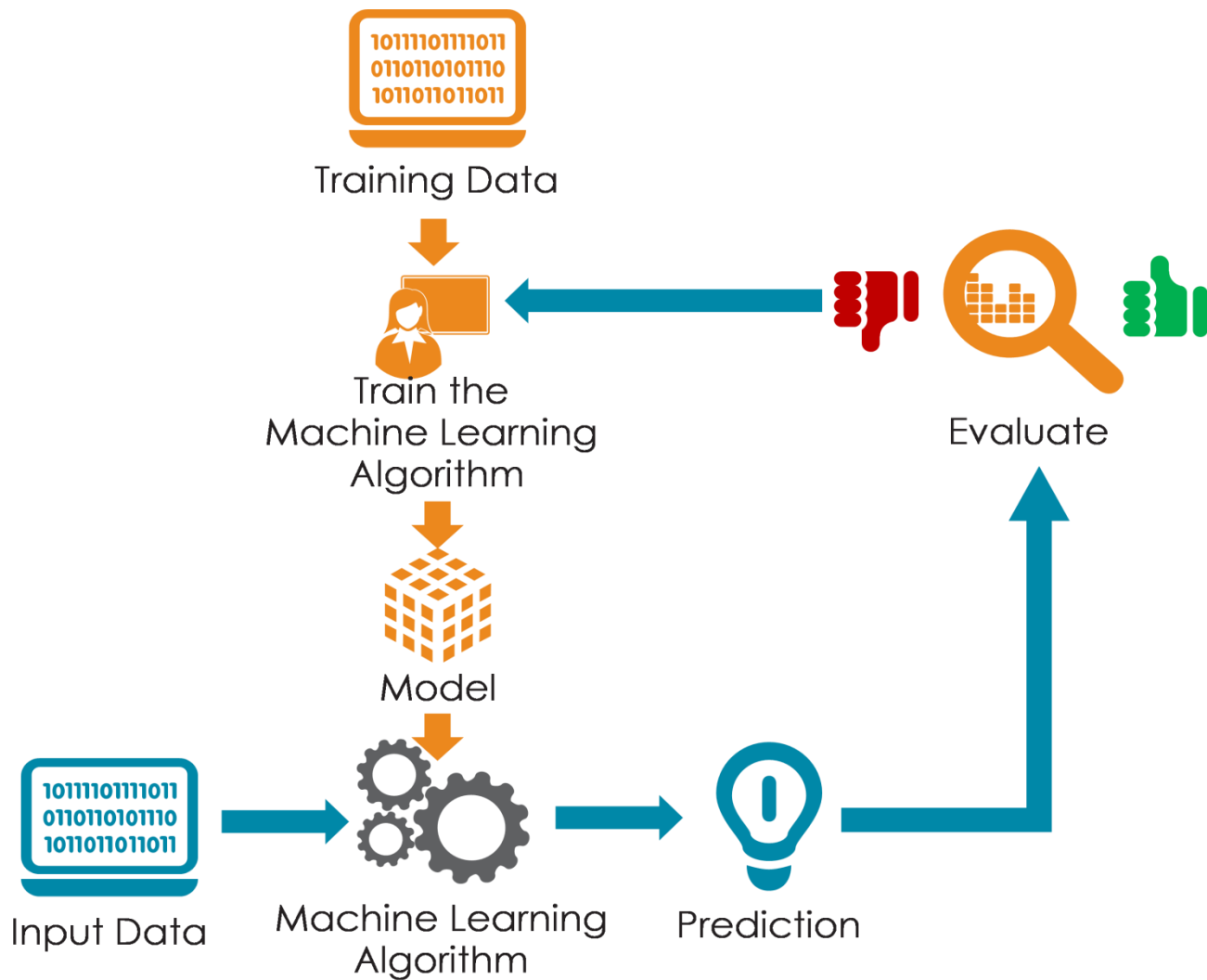


Feedback and
Adaptation

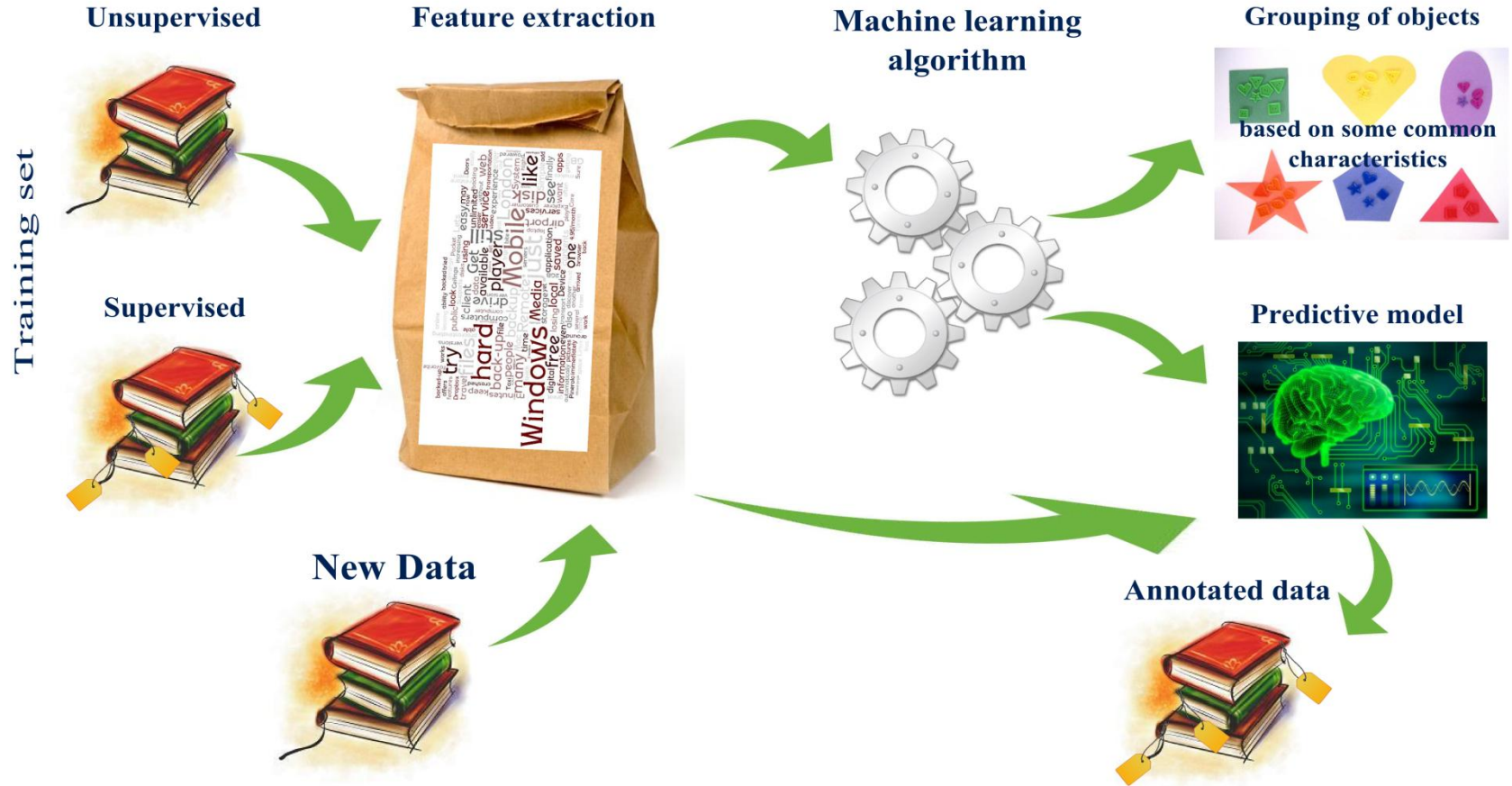




The pattern recognition system



Machine learning workflow



Terminology

- data set: $D = \{x_1, x_2, \dots, x_n\}$
- Instance/sample
- attribute/feature (space)
- feature vector: $x_i = (x_{i1}; x_{i2}; \dots; x_{id})$ d: dimension
- learning/training (sample/set)/testing
- Hypothesis, ground-truth, learner

Terminology

- Prediction, example $(\mathbf{x}_i, y_i)$, label, label space
- Classification: discrete value
- Regression: continuous value
- Binary classification: positive + negative
- Clustering
- Generalization, distribution, i.i.d.

Hypothesis Space

- Induction: generalization, concept (learning)
- Deduction: specialization
- ***Paradox***: large hypothesis space vs. limited training set (there exists one of multiple possible hypotheses consistent with training set, i.e., version space)

Empirical Error and Over-fitting

- **Error rate:** $\# \text{ of misclassified samples} / \# \text{ of total samples}$
- **Accuracy:** $1 - \text{error rate}$
- **Error:** the difference between prediction and ground truth
- Training error a.k.a. empirical error
- Test error a.k.a. generalization error
- Overfitting vs. underfitting

Overfitting vs. Underfitting

树叶训练样本



新样本



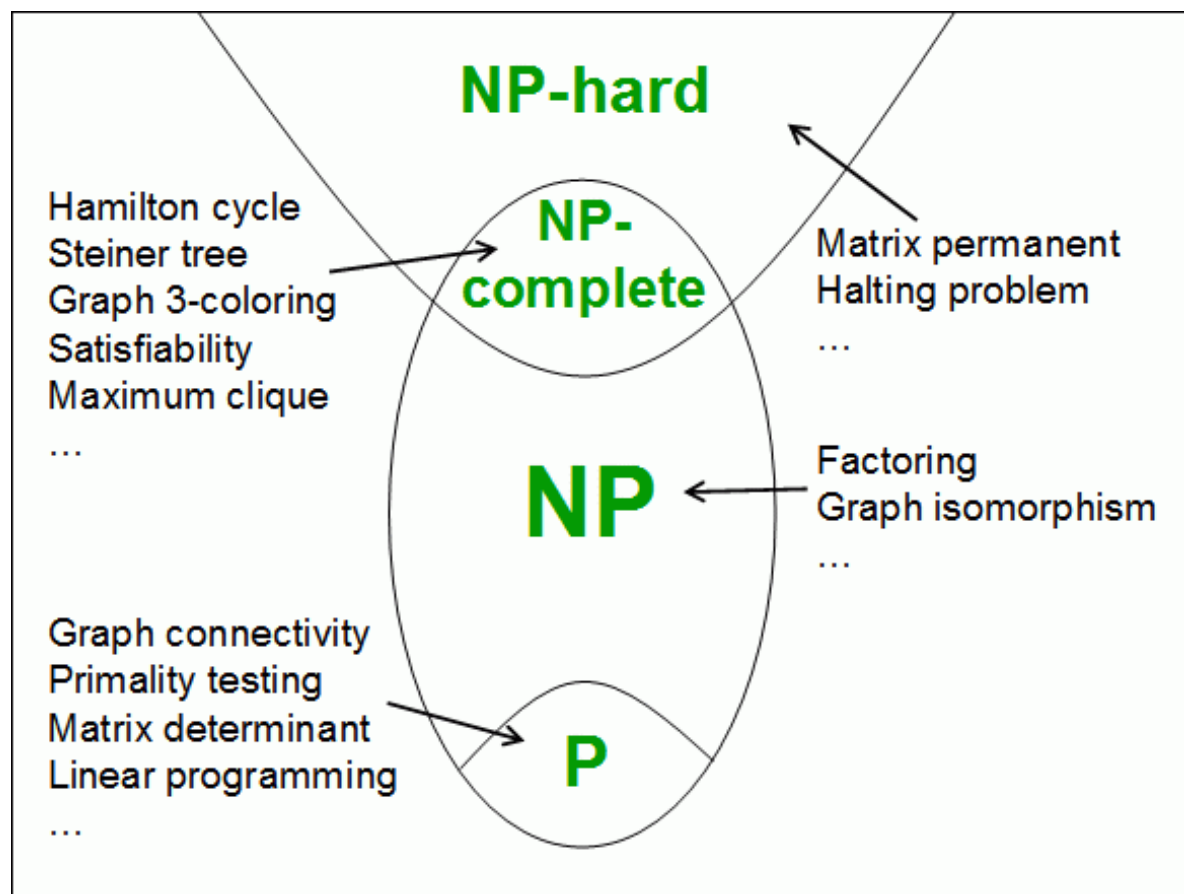
过拟合模型分类结果：
→ 不是树叶
(误以为树叶必须有锯齿)



欠拟合模型分类结果：
→ 是树叶
(误以为绿色的都是树叶)

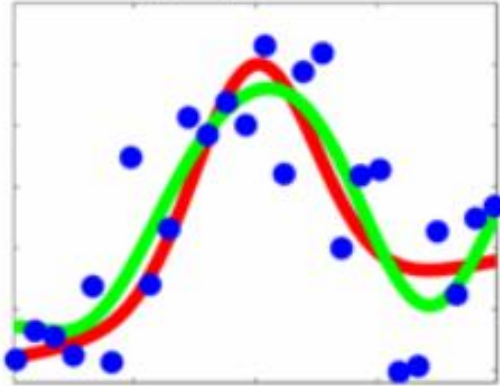
Problem Computability

- ML problems are usually NP-hard.
- Learning algorithms are required to be accomplished in polynomial time.
- Overfitting is unavoidable.
- $P \neq NP$
- ***Paradox***: if overfitting can be avoided, then the optimal solution is attainable via minimizing empirical error. It leads to $P = NP$.

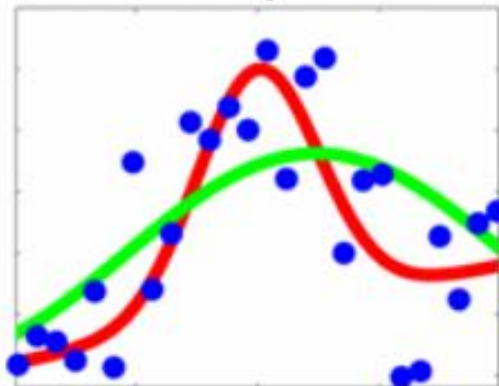


Model Selection

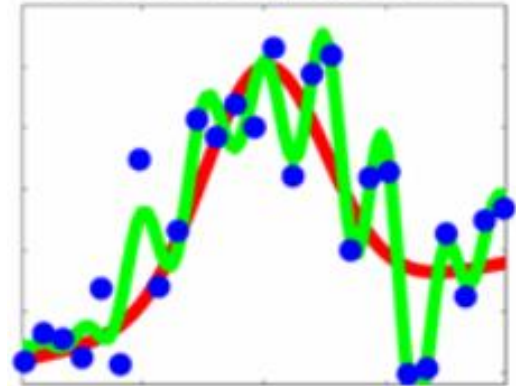
Learned function
with appropriate model



Learned function
with too simple model



Learned function
with too complex model

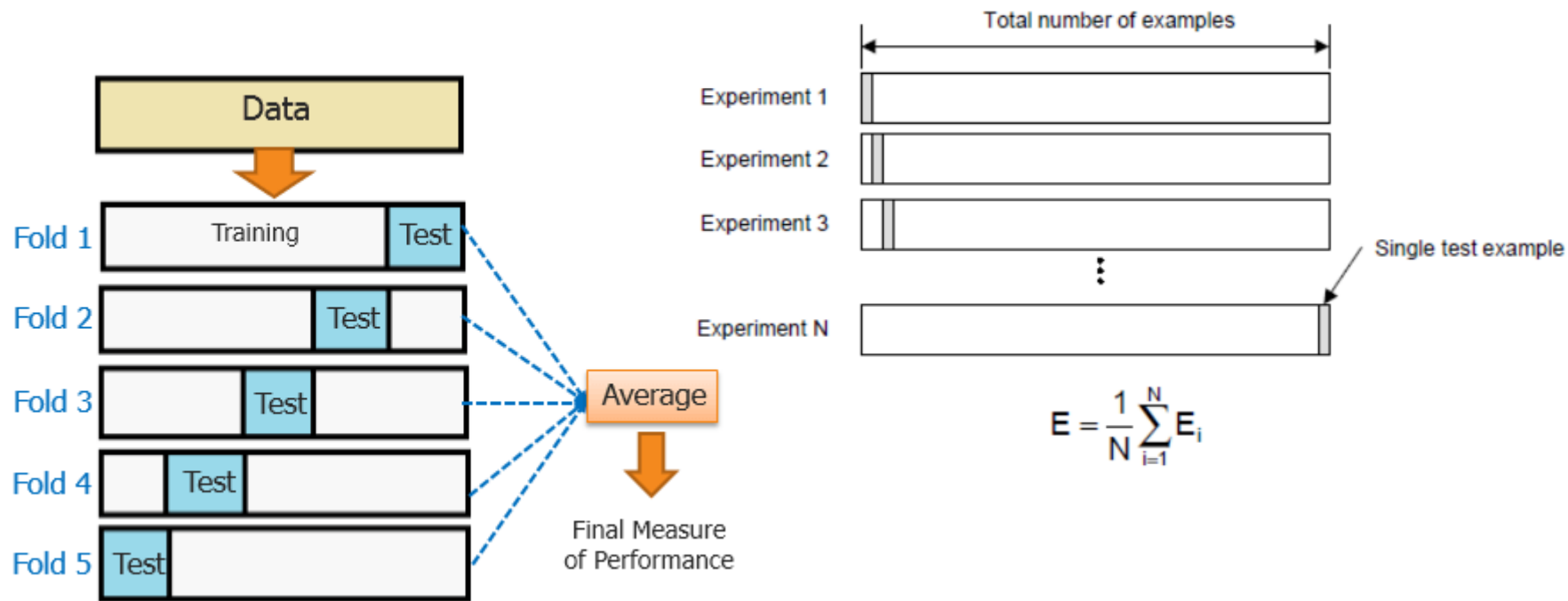


Evaluation

- Testing error: approximate generalization error
- $D = D_{\text{train}} + D_{\text{test}}$ || $D = D_{\text{train}} + D_{\text{val}} + D_{\text{test}}$
- Hold-out: $D_{\text{train}} \cap D_{\text{test}} = \emptyset$ (Stratified Sampling + Average)
- Cross-Validation: k -fold CV (LOO: Leave-One-Out)
- Bootstrapping: sampling with replacement

$$\lim_{m \rightarrow \infty} \left(1 - \frac{1}{m}\right)^m \mapsto \frac{1}{e} \approx 0.368$$

Cross-Validation



Parameter Tuning

- Hyper-parameter: empirical value, e.g., regularizer
- Model parameter: learnable, e.g., CNN
- Grid-searching
- Large complexity
- Examined on validation set

Performance Measure

- Regression: MSE(Mean Squared Error)

$$E(f; D) = \frac{1}{m} \sum_{i=1}^m (f(\mathbf{x}_i) - y_i)^2$$

Data set D
Learner f

$$E(f; \mathcal{D}) = \int_{\mathbf{x} \sim \mathcal{D}} (f(\mathbf{x}) - y)^2 p(\mathbf{x}) d\mathbf{x}$$

Data distribution
PDF $p(\mathbf{x})$

Error Rate vs. Accuracy

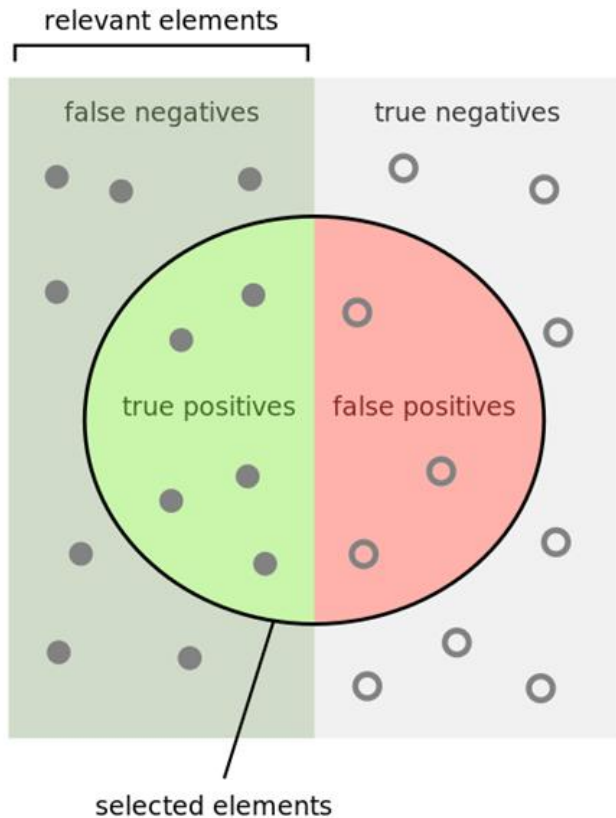
$$E(f; D) = \frac{1}{m} \sum_{i=1}^m \mathbb{I}(f(\mathbf{x}_i) \neq y_i)$$

$$\begin{aligned} \text{acc}(f; D) &= \frac{1}{m} \sum_{i=1}^m \mathbb{I}(f(\mathbf{x}_i) = y_i) \\ &= 1 - E(f; D) . \end{aligned}$$

$$E(f; \mathcal{D}) = \int_{\mathbf{x} \sim \mathcal{D}} \mathbb{I}(f(\mathbf{x}) \neq y) p(\mathbf{x}) d\mathbf{x}$$

$$\begin{aligned} \text{acc}(f; \mathcal{D}) &= \int_{\mathbf{x} \sim \mathcal{D}} \mathbb{I}(f(\mathbf{x}) = y) p(\mathbf{x}) d\mathbf{x} \\ &= 1 - E(f; \mathcal{D}) . \end{aligned}$$

Precision-Recall



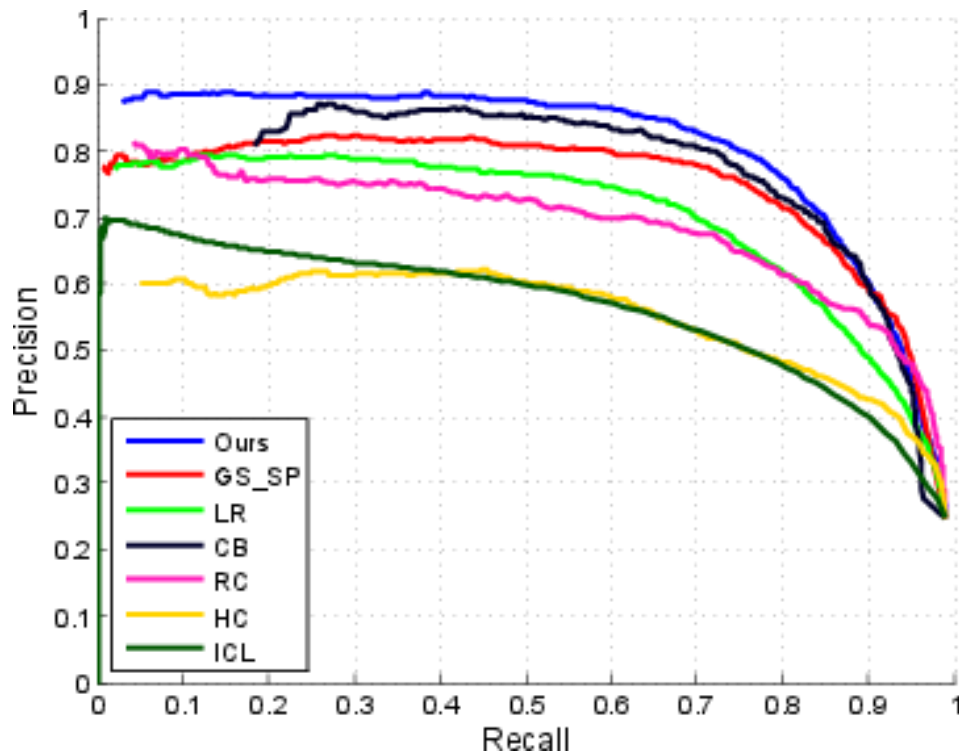
How many selected items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are selected?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$

P-R Curve



Confusion Matrix

$$\text{macro-}P = \frac{1}{n} \sum_{i=1}^n P_i ,$$

$$\text{macro-}R = \frac{1}{n} \sum_{i=1}^n R_i ,$$

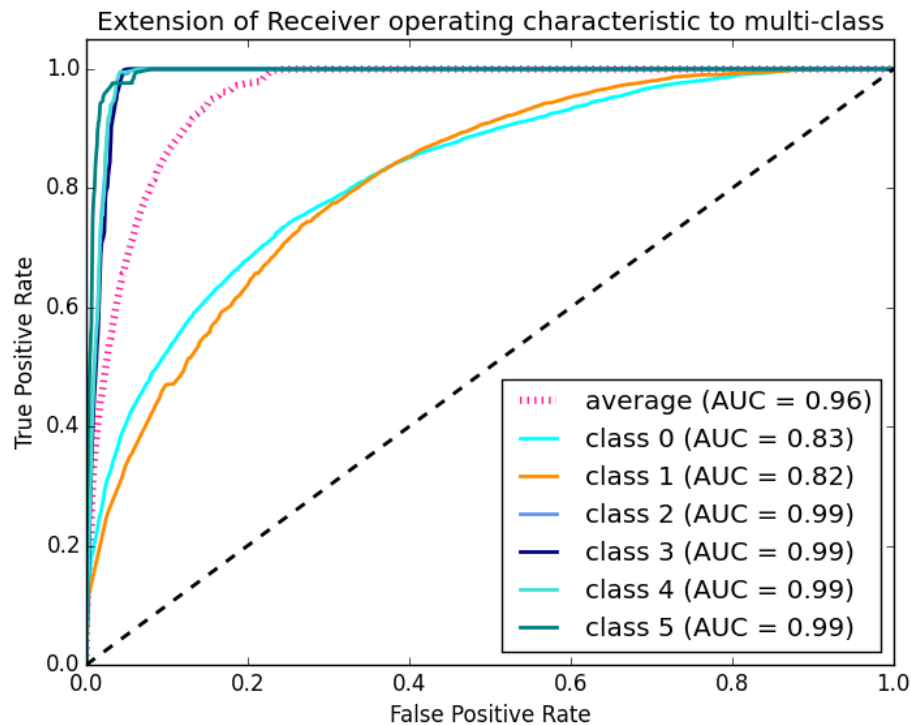
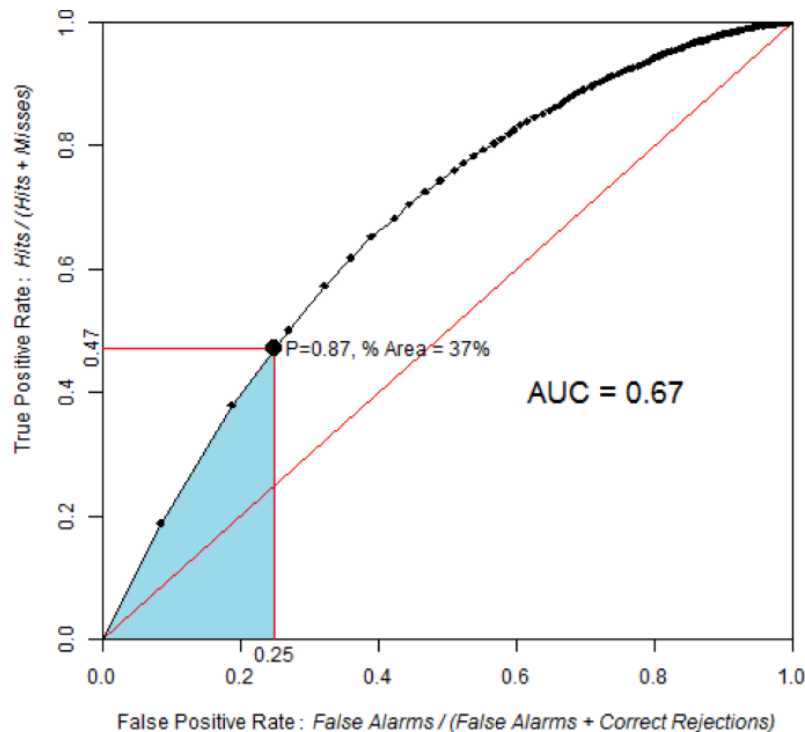
$$\text{macro-}F1 = \frac{2 \times \text{macro-}P \times \text{macro-}R}{\text{macro-}P + \text{macro-}R} .$$

$$\text{micro-}P = \frac{\overline{TP}}{\overline{TP} + \overline{FP}} ,$$

$$\text{micro-}R = \frac{\overline{TP}}{\overline{TP} + \overline{FN}} ,$$

$$\text{micro-}F1 = \frac{2 \times \text{micro-}P \times \text{micro-}R}{\text{micro-}P + \text{micro-}R} .$$

ROC and AUC



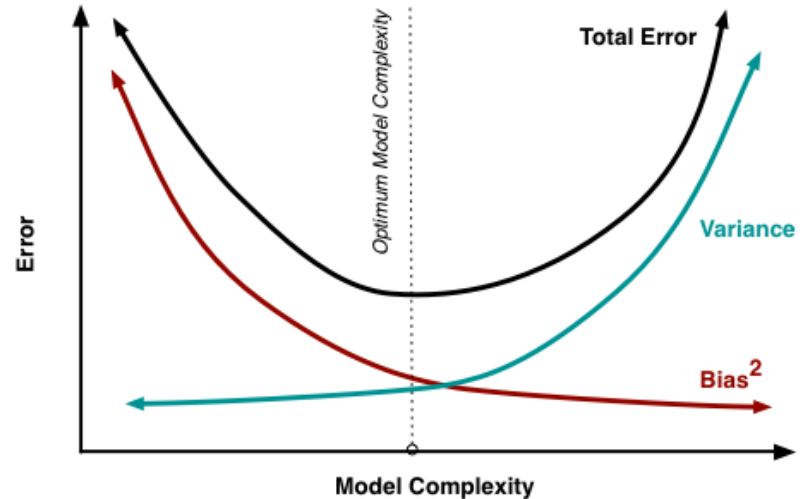
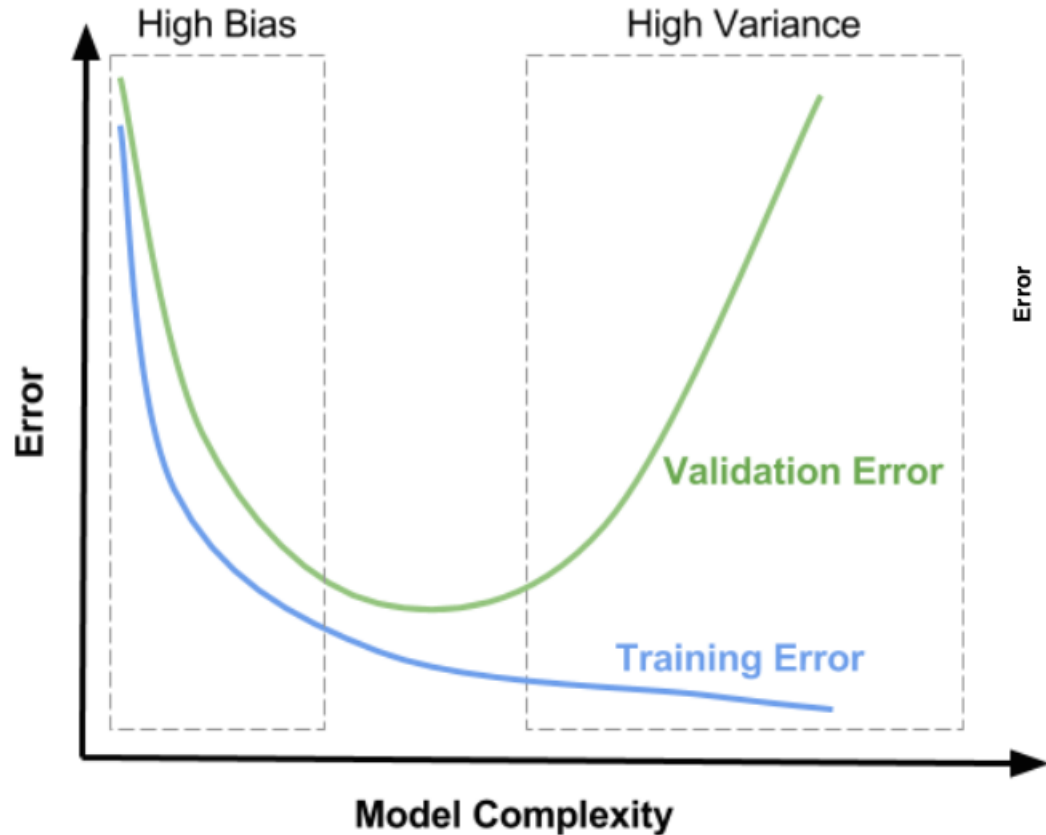
Cost-Sensitive Classification

Contingency Table

	Actual Positive $y_i = 1$	Actual Negative $y_i = 0$
Predicted Positive $c_i = 1$	C_{TP_i}	C_{FP_i}
Predicted Negative $c_i = 0$	C_{FN_i}	C_{TN_i}

$$Cost(f(S)) = \sum_{i=1}^N y_i (c_i C_{TP_i} + (1 - c_i) C_{FN_i}) + (1 - y_i) (c_i C_{FP_i} + (1 - c_i) C_{TN_i})$$

Bias-Variance Tradeoff



$$E(f; D) = \text{bias}^2(\mathbf{x}) + \text{var}(\mathbf{x}) + \varepsilon^2$$

Expected Generalization Error

Mathematics

- Mathematical Foundations of Machine Learning
- <https://voices.uchicago.edu/willett/teaching/fall-2019-mathematical-foundations-of-machine-learning/>
- 2019 Mathematics of Machine Learning Summer School
- <https://www.youtube.com/watch?v=3wbLr-NnIKI&list=PLTPQEx-31JXhguCush5J7OGnEORofoCW9>

Reference

- <https://viso.ai/deep-learning/pattern-recognition/>
- How artificial intelligence is changing our society
<https://www.youtube.com/watch?v=-ePZ7OdY-Dw>